BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, BANNU

INTERMEDIATE (12TH CLASS) EXAMINATION, 2021

MATHEMATICS

Past Paper 2021

Retyped faithfully from the original board paper — same questions, no changes.

SECTION–A

Time Allowed: 20 Minutes | **Maximum Marks:** 20

Note: CAVEAT: ilmkidunya's 2021 objective page serves a 'Model Paper / Mathematics Paper – XII' scan (not the actual 2021 objective print). Transcribed as served.

Q. The range of y = 1/(x−3) is .....

(A) R
(B) R−{y|y=3}
(C) R−{y|y=0}
(D) R−{3}

Q. The function f(x) = |x|/x is discontinuous at..

(A) x = 1
(B) x = −1
(C) x = 0
(D) x = 2

Q. d/dx (3^x) = ...

(A) x3^{x−1}
(B) 3^x ln 3
(C) 3^x
(D) 3^{x+1}

Q. d/dx (Sinh⁻¹3x) = ...

(A) 3/√(1+9x²)
(B) 3/√(1−9x²)
(C) −3/√(1+9x²)
(D) −3/√(1−9x²)

Q. f(x) = 2^{3x} then f'(x) = ...

(A) 3^5 2^{3x}
(B) 3^5 2^{3x}log2
(C) 3^5 2^{3x}log2^5
(D) 3^5 2^{3x}(log2)^5

Q. The slope of the tangent line on a curve y = f(x) at a particular point P(x1,y1) is ...

(A) f'(x1)
(B) f'(y1)
(C) f'(x1,y1)
(D) f(x1,y1)

Q. A function f(x) is defined at a number K and either f'(K) = 0 or f'(K) does not exist. Then the number K is called a .....

(A) Maximum value
(B) Minimum value
(C) Stationary value
(D) Critical value

Q. The vector function f(x) = <Sin x, $(1-x^2)^{-1}$, lnx> is continuous at ...

(A) R
(B) $x \neq 1$
(C) $R-\{0\}$
(D) $x>0, x \neq 1$

Q. $\int 5^{2x} dx$ = ...

(A) $5^{2x}/2 + c$
(B) $5^{2x}/(2\ln 5) + c$
(C) $5^{2x}/\ln 5 + c$
(D) $(2.5^{2x})/\ln 2 + c$

Q. The eccentricity of an ellips lies between.....

(A) −1 and [illegible in source scan]
(B) 0 and 1
(C) 1 and 2
(D) None of these

Q. In $\int dx/(a^2+x^2)\, dx$ we substitute x = ...

(A) Tan θ
(B) Sec θ
(C) 3 Tan θ
(D) 3 Sec θ

Q. In $\int dx/((x^2+1)(x^2-1))$ we use the method of .....

(A) By Parts
(B) Logarithmic
(C) Partial Fraction
(D) None of these

Q. $\int_1^2 (x^2+2)dx$ = ...

(A) 12
(B) 6
(C) 2
(D) 0

Q. The slope of a straight line coincides or parallel to X-axis is .....

(A) 1
(B) 0
(C) −1
(D) ∞

Q. When the radius of a circle is zero then such a circle is called .....circle.

(A) Semi
(B) Point
(C) Virtual
(D) Imaginary

Q. The slope of a line bisecting the 1st and 3rd quadrant is .....

(A) 0
(B) 1
(C) π/4
(D) −1

Q. In parabola $y^2-12x = 0$ the length of focal chord is .....

(A) 12 (B) −12

(C) 3/4 (D) 4/3

Q. The equation of normal to the elips $x^2/a^2+y^2/b^2 = 1$ at a point P(x1, y1) is .....

(A) $xx1/a^2 + yy1/b^2 = 1$ (B) $x1/x + y1/y = 1$

(C) $(y-y1)/(y1/b^2) = (x-x1)/(x1/a^2)$ (D) $xx1/a^2 - yy1/b^2 = 1$

Q. The differential equation $d^3y/dx^3 - 9(d^2y/dx^2)^2 + x^4(dy/dx)^3 + 2y + 3$ is of degree,

(A) 1 (B) 2

(C) 3 (D) None of these

Q. If the function values f(a) and f(c) at x=a and x=c have opposite signs then the root lies in the interval.....

(A) [c,a] (B) [a,c]

(C) (c,a) (D) (a,c)

Section – B

Maximum Marks: 50

Note: MATHEMATICS (New), Inter Part-II (Fresh/Reappear). Time allowed for Section – B and Section – C is 2 Hours and 40 minutes. No paper code/time header on the subjective scan (ALP format).

QII. (Marks 50) Answer any TEN parts. Each part carries FIVE marks.

1. Find the composite function f[g(x)] and g[f(x)] for the functions $f(x) = 2x^2+9$ and $g(x) = \sqrt{(x+3)}$

2. Evaluate $\text{Limit}(x\to1)\ (x^2+2x-3)/(x-1)$

3. Use first principle rule to determine the derivative of $f(x) = 1/\sqrt{(2x+3)}$

4. Use any suitable rule to perform dy/dx for the function $y = (1+\text{Cos}^2 2x)/\text{Cosec}3x$

5. Find an equation of the tangent line to the curve $x^2+y^2 = 13$ at $(2,-3)$

6. Find F''(0) if $F(\theta) = \text{Sin}^2\theta i+\text{Cos}2\theta j+\theta^3 k$

7. Evaluate the integral $\int x\ \text{Tan}^{-1}x\ dx$

8. Evaluate $\int(0 \text{ to } 1)\ 3x\sqrt{(3x^2+2)}\ dx$

9. Find the equation of the line that passes through the pair of points A(3,4) and B(−5,5)

10. Find the condition at which the line $ax+by+c = 0$ touches the circle $x^2+y^2+2gx+2fy+c = 0$

11. Find the points of intersection in between the line $2x-y+1 = 0$ and the parabola $y^2 = 9x$

12. Solve the differential equation $dy/dt = y/t + t^2/y^2$

13. Verify Euler's theorem for the homogenous function $u = f(x,y) = ax^2+2hxy+by^2$

Section – C

Maximum Marks: 30

Note: Note: Attempt any THREE questions. Each question carries equal marks.

QIII. Q-III

(a) Find iterate x_3 of Newton−Raphson iterative method for the function $f(x) = x^3-1$, $x_0 = 1$

(b) For what value of C the line $y = x+c$ will touch the hyperbola $x^2/25 - y^2/16 = 1$

QIV. Q-IV

(a) Find the equation of the circle, which contains the points (0,0), (0,3) and (−4,0)

(b) Find the area of the triangular region whose vertices are A(2,4) B(−2,2) & C(0,0)

QV. Q-V

(a) Find the critical values of the function $f(x) = 2x^3-3x^2-60x+5$

(b) Find d^2y/dx^2 for the parametric functions x(z), y(z), when $x = 4z^2+1$, $y = 6z^3+1$

QVI. Q-VI

(a) Evaluate $\int dx/(x^2-1)$

(b) Find dy/dx when $x^2y^2+2y^2 = x^2+2$ implicitly.